

PROGRAM

;Segment defined as stack segment and named as "STSEG"

;It can be named with programmer's choice but must not be a keyword

STSEG SEGMENT

DB 64 DUP(?) ;defining an array of 64bytes, DB (Defining Byte) and DUP (Duplicate)
STSEG ENDS ;stack segment ends here, ENDS (End Segment)

;Segment defined as data segment and named as "DTSEG"

;It can be named with programmer's choice but must not be a keyword

DTSEG SEGMENT

DATA1 DB 32 ;Declaring a variable "DATA1" with data type byte (DB) and initializing with value 32 decimal
DATA2 DB 21 ;Declaring a variable "DATA2" with data type byte (DB) and initializing with value 21 decimal
DATA3 DB 12 ;Declaring a variable "DATA3" with data type byte (DB) and initializing with value 12 decimal
DATA4 DB 23 ;Declaring a variable "DATA4" with data type byte (DB) and initializing with value 23 decimal
DATA5 DB 39 ;Declaring a variable "DATA5" with data type byte (DB) and initializing with value 39 decimal
SUM DB 0 ;Declaring a variable "SUM" with data type byte (DB) and initializing with value 0 decimal

DTSEG ENDS ;data segment ends here, ENDS (End Segment)

;Segment defined as code segment and named as "CDSEG"

;It can be named with programmer's choice but must not be a keyword

CDSEG SEGMENT

MAIN PROC ;main procedure starts here (similar to void main (void))

;informing the code about assumptions that STSEG is SS, DTSEG is DS, and ;CDSEG is CS using
;ASSUME directive

ASSUME SS:STSEG, DS:DTSEG, CS:CDSEG

MOV AX, DTSEG ;initializing the data segment by moving the value of DTSEG in DS
MOV DS, AX

MOV AL, DATA1 ;AL = 32
ADD AL, DATA2 ;AL = AL + DATA2
ADD AL, DATA3 ;AL = AL + DATA3
ADD AL, DATA4 ;AL = AL + DATA4
ADD AL, DATA5 ;AL = AL + DATA5

MOV SUM, AL ;SUM = AL

;transferring the control to the operating system by using interrupt 21h

MOV AH, 4CH ;AH = 4C (hexadecimal)

INT 21H

MAIN ENDP ;main procedures ends here, ENDP (End Procedure)

CDSEG ENDS ;code segment ends here, ENDS (End Segment)

END MAIN ;the whole code (program) ends here

Assembling and Linking the program to create Executable file

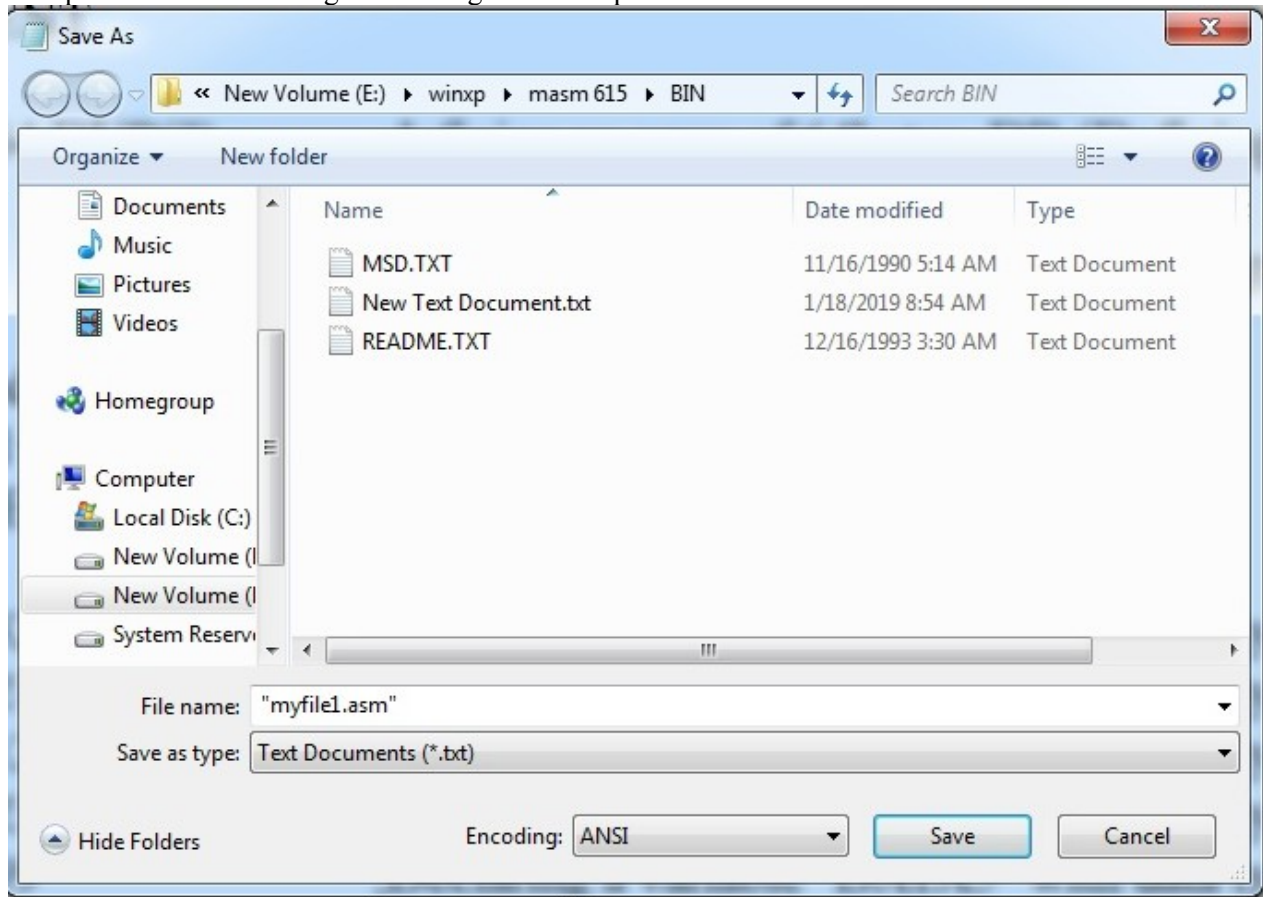
Download and extract MASM615 zip file from course website in the drive E.

In the same drive E, copy **winxp** folder.

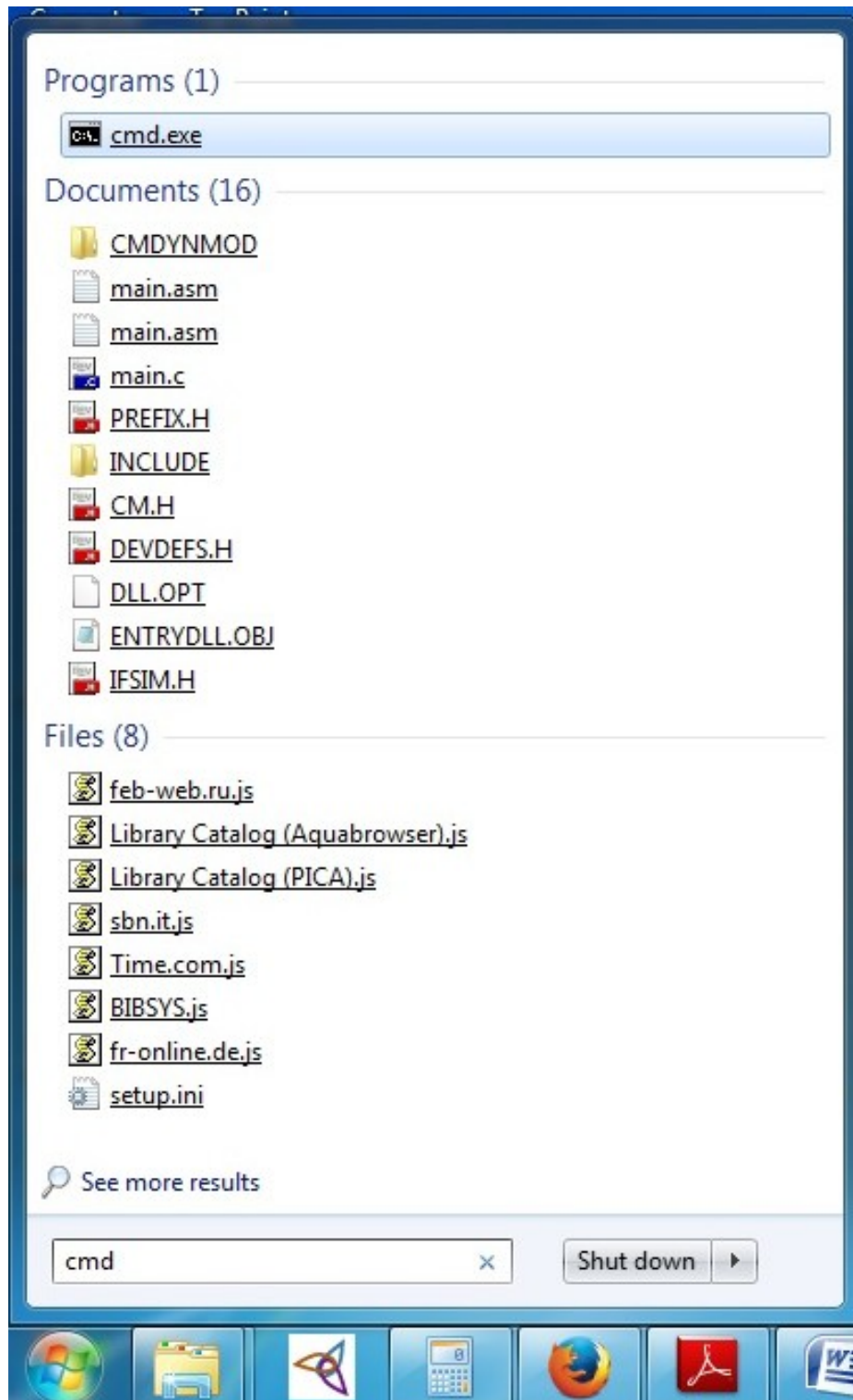
Write the program using NOTEPAD or any other EDITOR software

Store it as myfile.asm, in the BIN folder under MASM615.

Use quotations when storing the file as given in the picture below



Go to command line window.



cmd window appears. Do as describe in the following figure.

```
Administrator: C:\Windows\system32\CMD.exe
C:\Users\SHAKIR>E:
E:\>CD WINXP                                cd for changing directory
E:\winxp>DIR /AD /P                          takin all directory page wise
Volume in drive E is New Volume
Volume Serial Number is 4A90-933A

Directory of E:\winxp
12/19/2019  01:10 PM    <DIR>          .
12/19/2019  01:10 PM    <DIR>          ..
01/17/2019  12:33 PM    <DIR>          masm 615      masm 615 folder
               0 File(s)                0 bytes
               3 Dir(s)  122,793,209,856 bytes free

E:\winxp>CD MASM 615                          chang driectory to masm 615
E:\winxp\masm 615>CD BIN                      and go to bin
E:\winxp\masm 615\BIN>DIR MYFILE1.ASM
Volume in drive E is New Volume
Volume Serial Number is 4A90-933A

Directory of E:\winxp\masm 615\BIN
12/19/2019  01:20 PM                2,051 myfile1.asm
               1 File(s)                2,051 bytes
               0 Dir(s)  122,793,209,856 bytes free

E:\winxp\masm 615\BIN>MASM MYFILE1.ASM        use masm to assemble
Microsoft (R) MASM Compatibility Driver
Copyright (C) Microsoft Corp 1993. All rights reserved.

Invoking: ML.EXE /I. /Zm /c /Ta MYFILE1.ASM

Microsoft (R) Macro Assembler Version 6.14.8444
Copyright (C) Microsoft Corp 1981-1997. All rights reserved.

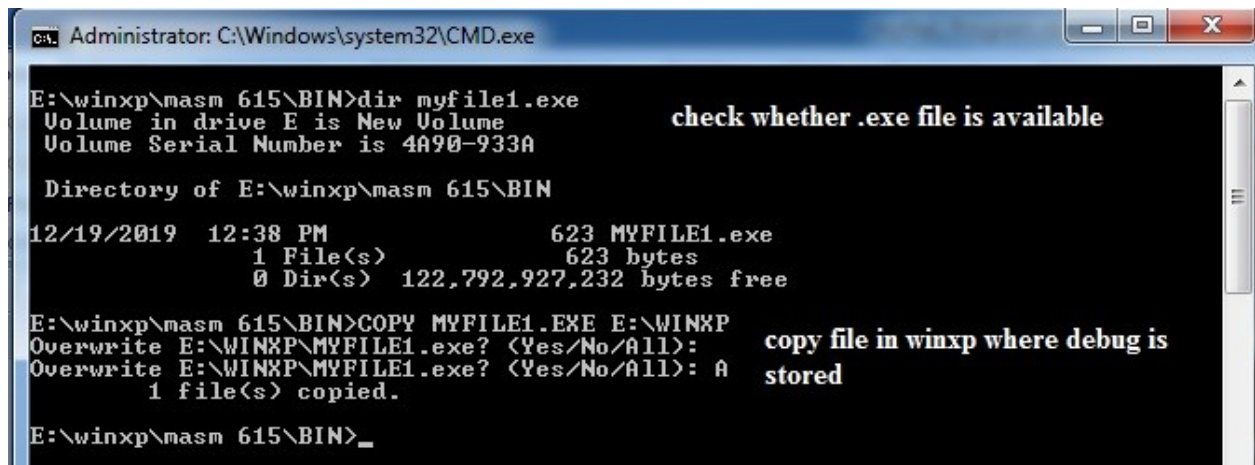
Assembling: MYFILE1.ASM

E:\winxp\masm 615\BIN>LINK MYFILE.OBJ        use link to make .exe file
Microsoft (R) Segmented Executable Linker Version 5.60.339 Dec  5 1994
Copyright (C) Microsoft Corp 1984-1993. All rights reserved.

Run File [MYFILE.exe]:
List File [nul.map]:
Libraries [.lib]:
Definitions File [nul.def]:
LINK : warning L4021: no stack segment

E:\winxp\masm 615\BIN>
```

Copy the .exe file just created in the folder where debug utility is stored. It is described in the following figure.



A screenshot of a Windows Command Prompt window titled "Administrator: C:\Windows\system32\CMD.exe". The prompt is at the directory E:\winxp\masm 615\BIN. The user enters 'dir myfile1.exe', and the output shows a file named MYFILE1.exe, 623 bytes in size. The user then enters 'COPY MYFILE1.EXE E:\WINXP', and the prompt asks for confirmation to overwrite the file. The user enters 'A', and the output shows '1 file(s) copied.'.

```
Administrator: C:\Windows\system32\CMD.exe

E:\winxp\masm 615\BIN>dir myfile1.exe
Volume in drive E is New Volume
Volume Serial Number is 4A90-933A

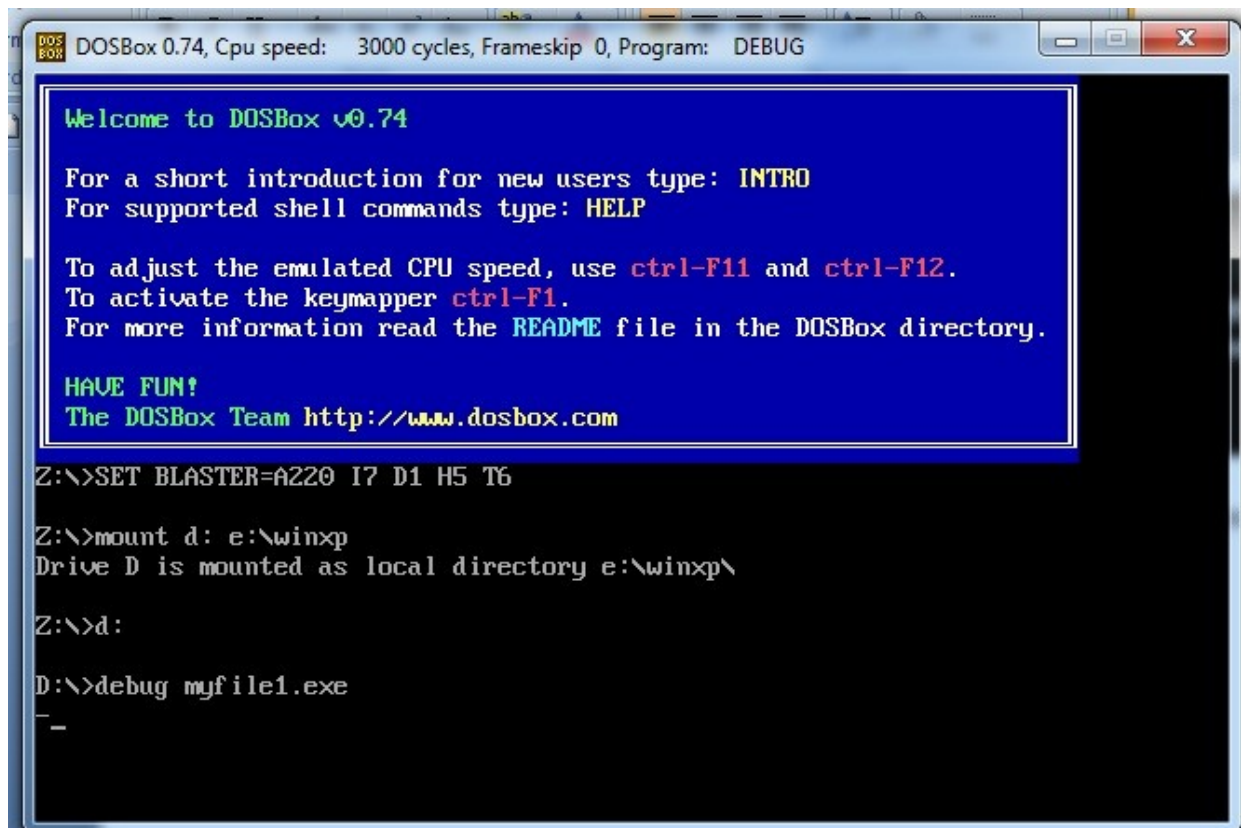
Directory of E:\winxp\masm 615\BIN

12/19/2019  12:38 PM                623 MYFILE1.exe
               1 File(s)                623 bytes
               0 Dir(s)  122,792,927,232 bytes free

E:\winxp\masm 615\BIN>COPY MYFILE1.EXE E:\WINXP
Overwrite E:\WINXP\MYFILE1.exe? <Yes/No/All>:
Overwrite E:\WINXP\MYFILE1.exe? <Yes/No/All>: A
               1 file(s) copied.

E:\winxp\masm 615\BIN>
```

Now, open DOSBox, mount and open the .exe file in debug. The process is shown in the figure below.



A screenshot of a DOSBox window titled "DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG". The window shows a welcome message and instructions for new users. The user enters 'SET BLASTER=A220 I7 D1 H5 T6', then 'mount d: e:\winxp', and 'debug myfile1.exe'.

```
DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG

Welcome to DOSBox v0.74

For a short introduction for new users type: INTRO
For supported shell commands type: HELP

To adjust the emulated CPU speed, use ctrl-F11 and ctrl-F12.
To activate the keymapper ctrl-F1.
For more information read the README file in the DOSBox directory.

HAVE FUN!
The DOSBox Team http://www.dosbox.com

Z:\>SET BLASTER=A220 I7 D1 H5 T6

Z:\>mount d: e:\winxp
Drive D is mounted as local directory e:\winxp\

Z:\>d:

D:\>debug myfile1.exe
_
```

Use the extension “.exe” is mandatory.

Further, use unassembled command to analyze the program.

```
DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG

HAVE FUN!
The DOSBox Team http://www.dosbox.com

Z:\>SET BLASTER=A220 I7 D1 H5 T6

Z:\>mount d: e:\winxp
Drive D is mounted as local directory e:\winxp\

Z:\>d:

D:\>debug myfile1.exe
-u
076F:0000 B86E07      MOV     AX,076E  DTSEG is replaced by its value
076F:0003 8ED8        MOV     DS,AX    note the value 076E
076F:0005 A00000      MOV     AL,[0000] DATA1 is replaced by its address 0000h
076F:0008 02060100    ADD     AL,[0001] DATA2 is replaced by its address 0001h
076F:000C 02060200    ADD     AL,[0002] DATA3 is replaced by its address 0002h
076F:0010 02060300    ADD     AL,[0003] DATA4 is replaced by its address 0003h
076F:0014 02060400    ADD     AL,[0004] DATA5 is replaced by its address 0004h
076F:0018 A20500      MOV     [0005],AL SUM is replaced by its address 0005h
076F:001B B44C        MOV     AH,4C
076F:001D CD21      INT     21      Program ends at 001D
076F:001F 0000      ADD     [BX+SI],AL
```

Data Segment value is 076E.

The variable DATA1 is located on the memory location 076E:0000

The variable DATA2 is located on the memory location 076E:0001

The variable DATA3 is located on the memory location 076E:0002

The variable DATA4 is located on the memory location 076E:0003

The variable DATA5 is located on the memory location 076E:0004

The variable SUM is located on the memory location 076E:0005

To check the data values stored in data segment write the command

- D 076e:0000 0007

As discussed in the following figure.

```
DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
Drive D is mounted as local directory e:\winxp\
Z:\>d:
D:\>debug myfile1.exe
-u
076F:0000 B86E07      MOV     AX,076E      starting IP is 0000h
076F:0003 8ED8        MOV     DS,AX
076F:0005 A00000      MOV     AL,[0000]
076F:0008 02060100    ADD     AL,[0001]
076F:000C 02060200    ADD     AL,[0002]
076F:0010 02060300    ADD     AL,[0003]
076F:0014 02060400    ADD     AL,[0004]
076F:0018 A20500      MOV     [0005],AL
076F:001B B44C        MOV     AH,4C
076F:001D CD21      INT     21
076F:001F 0000      ADD     [BX+SI],AL      ending IP is 001Fh
-d076e:0000 0007
076E:0000 20 15 0C 17 27 00 00 00      data variable values ...'...
-g=0000 001f                          executing the program using g command
Program terminated normally
-d076e:0000 0007
076E:0000 20 15 0C 17 27 7F 00 00      showing the result ...'...
_
```

The sum is 7Fh.